



Enzyme-DNA chimeras: Construction, allostery, applications

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Contents

1. Introduction	258
2. Enzyme-DNA chimera	259
2.1 Basic principle of the method	259
2.2 Examples of different enzymes modulated by the enzyme-DNA method	261
3. Principles of designing enzyme-DNA chimeras	265
3.1 Selection of the enzyme and DNA attachment sites	265
3.2 Design of the sequence of the DNA arms	266
3.3 Two-step chimera construction	267
4. Design and synthesize enzyme-DNA chimeras	268
4.1 Equipment	268
4.2 Chemicals and materials	269
4.3 Protocols	269
4.4 Example of protocol—RLuc-DNA chimera synthesis	273
4.5 Troubleshooting	275
5. Discussion	276
5.1 Selection of target enzyme: Possibilities and challenges	276
5.2 Quantitative characterization of the mechanical perturbation	277
5.3 Applications: Enzyme-DNA chimeras as molecular probes	278
6. Summary	279
Acknowledgments	279
References	279

Abstract

We describe the operational principle, synthesis, and applications of the enzyme-DNA chimeras. These are supramolecular constructions where a DNA spring is coupled to an enzyme and introduces artificial allosteric control of the enzyme. This method is universal and can be applied to various enzymes and proteins. In addition, this method is

versatile as the stresses applied by the DNA spring on the enzymes can be fine-tuned semi-continuously and thus their enzymatic activities can be modulated gradually. We give detailed protocols for the synthesis of these molecules. Summarizing our experience with different enzymes, we explain their use for fundamental studies of conformational plasticity, as well as the potential as molecular probes.



1. Introduction

The living cell is all about functional supra-molecular complexes; in contrast, our ability to imagine and produce such constructions artificially is at present quite limited. The purpose of this chapter is to introduce, in a practical way, a family of artificial enzyme–DNA complexes: the principle of their function, some applications, and most importantly the detailed protocols we use for their synthesis. Our objective is to explain what sort of interesting questions might be pursued through these constructions and to put these tools in the hands of whoever may want to use them or build upon them. Most of what we report here has appeared in print before, but the material is scattered through several publications, the more practical details often hidden in the “supplementary material” sections. Here we aim to present a more unified view of the subject matter.

In order to understand the working principle of the enzyme–DNA chimeras, it is useful to recapitulate the evolution of ideas concerning certain generic features of enzyme–substrate interactions, culminating in a generalized notion of allostery. Enzymes typically bind their substrates (reactants) through a large contact surface, i.e. through several weak bonds rather than one strong bond. The weak bonds may involve a combination of Van der Waals, electrostatic, and hydrophobic interactions, and hydrogen bonds. The large interaction surface is implicit in Fisher’s lock and key theory, the first successful attempt to explain the specificity of enzymes on the basis of generic (not enzyme specific) features. However, the enzyme–substrate interaction surface is not dissimilar from other internal interaction surfaces of the folded polypeptide chain, such as the surface between two adjacent elements of the secondary structure. Therefore one might expect that substrate binding will in general significantly perturb the structure of the protein. Koshland’s great discovery, the induced fit theory, was arrived at through rather more biochemical reasoning ([Koshland, 1995](#)), but it actually reconciles this state of affairs and the lock and key mechanism: substrate and enzyme do fit together snugly, in a deformed state of the enzyme stabilized by substrate binding. Here again, discoveries followed their own path, but

conceptually induced fit must lead to allostery, as vividly demonstrated in the recent evolutionary model of allostery by Tlsty and collaborators (Tlsty, Libchaber, & Eckmann, 2017). If binding of a ligand can deform the enzyme structure, two separate binding sites on the same enzyme will in general “talk” through the structure, so that binding of one ligand (the regulator) will affect binding and/or catalysis at the other site.

The explosion in structure determination over the past 50 years has transformed lock-and-key, induced fit, and allostery from biochemical theories to structure-based mechanisms (Fig. 1A). The beautiful structures of an enzyme in different conformations invite, as the basis for further experiments, simplified models which regard the molecule as a clockwork designed to move from conformation A to conformation B under the driving force of ligand binding. We may say that the clockwork provides a route from conformation A to conformation B, and, somewhat tautologically, that ligand binding stabilizes conformation B. Experiments may then consist of perturbing the clockwork through mutations to establish its components (Johnston, Whitney, Volkman, Doe, & Prehoda, 2011). However, as far back as the 1970s Frauenfelder and collaborators established experimentally that a protein in equilibrium explores a large number of conformational substates (Frauenfelder, Parak, & Young, 1988), a conclusion arrived at also since the earliest “all atom” simulations by Karplus and collaborators (Karplus & McCammon, 2002). More recent experiments have shown that enzymes are viscoelastic and can “flow” from one conformation to another (Alavi & Zocchi, 2018; Wang & Zocchi, 2011). The existence of a myriad conformational substates suggests that by applying a field of force, the enzyme can be reversibly deformed into many different structures, besides the ones resulting from ligand binding. If that is the case, a sufficient mechanical stress applied to the molecule will in general modulate the enzymatic activity, in other words, any field of force can in principle act as an allosteric regulator! The method of the DNA springs is based on this concept.



2. Enzyme-DNA chimera

2.1 Basic principle of the method

The enzyme-DNA chimeras are in vitro synthesized supra-molecular constructions consisting of an enzyme and a synthetic DNA oligomer of 40 to 60 bases (Fig. 1B). The two ends of the DNA are covalently attached through crosslinkers to two sites on the surface of the enzyme, away from

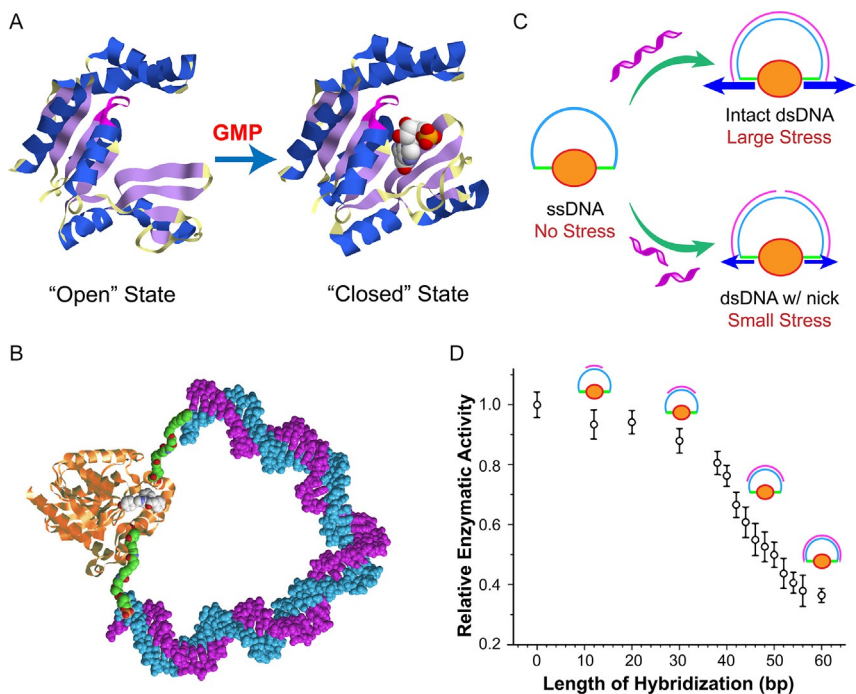


Fig. 1 (A) Example of structurally documented induced fit binding. The figure shows the structure of the enzyme Guanylate Kinase from yeast (PDB 1EX6 (apo enzyme) and 1EX7 (with GMP bound, shown in all atom representation)). The enzyme undergoes a large conformational change upon binding the substrate GMP. Namely, the two lobes of the protein swivel towards each other to form the “closed” state. The p-loop is colored in magenta. (B) Cartoon of an enzyme-DNA chimera. The enzyme is Renilla Luciferase (PDB 2PSJ), with the substrate coelenterazine-h shown in all atom representation. The DNA spring is cut out from the structure L8sa in the MD simulation of Lankas et al. (Lankaš, Lavery, & Maddocks, 2006), which shows a kink in the stressed DNA. The kink is at the origin of the softening transition mentioned in Section 5.2, so we believe this structure is a fair representation of the average conformation of the DNA in the enzyme-DNA chimeras. The DNA spring is shown attached at sites 161 and 188, through crosslinkers shown in green. The protein, DNA, and crosslinkers are drawn approximately to scale. (C) Schematic representation of how mechanical stress is controlled in the enzyme-DNA chimeras. With the DNA spring in the ss state there is essentially no stress on the enzyme. Adding the complementary DNA strand rigidifies the DNA, which acts as a leaf spring, pulling the attachment points on the enzyme apart (blue arrows). With a nick in the DNA spring, the stress on the enzyme is smaller. (D) The stress on the enzyme can be controlled semi-continuously by hybridizing the ss DNA of the chimera with complementary oligomers of different lengths. The plot shows the speed of the enzymatic reaction measured for a GK-DNA chimera with a 60mer DNA spring where the central portion is hybridized to oligomers of different lengths. The measurements are normalized to the activity of the ss chimera. Panel (B) Reprinted with permission from Ariyaratne, A., Wu, C., Tseng, C.-Y., & Zocchi, G. (2014). Dissipative dynamics of enzymes. *Physical Review Letters*, 113(19), 198101. doi:10.1103/PhysRevLett.113.198101. Copyright (2014) by the American Physical Society. Panel (D) Reproduced with permission from Choi, B., & Zocchi, G. (2007). Guanylate kinase, induced fit, and the allosteric spring probe. *Biophysical Journal*, 92(5), 1651–1658. doi:10.1529/biophysj.106.092866 via Copyright Clearance Center. Copyright (2007) by Elsevier.

the catalytic site so as not to impair function. In our constructions, the attachment sites on the enzyme are specific, defined by the reactive thiol groups of two Cysteines introduced by mutagenesis. The typical outcome (though there are exceptions) is that the enzyme-DNA chimera with the DNA in the single stranded (ss) form works just as well as the enzyme without DNA, in terms of catalytic activity. On the other hand, the ss DNA of the chimera can be hybridized with a complementary strand. This rigidifies the DNA, applying a stress on the enzyme. Namely, whereas a ss DNA molecule of 40–60 nucleotides (nt) is a very flexible polymer (the persistence length of ss DNA being around 3 bases or $\ell_{ss} \approx 1.5$ nm) (Smith, Cui, & Bustamante, 1996), the same length of ds DNA is a semi rigid rod (for ds DNA the persistence length is around 150 base pairs (bp) or $\ell_{ds} \approx 50$ nm) (Wang, Yin, Landick, Gelles, & Block, 1997). The ds DNA of the chimera therefore acts as a leaf spring (Fig. 1B), introducing a repulsive force between the attachment points, thus putting the enzyme under mechanical stress (Fig. 1C).

The virtue of the method is the external control of the force field: the experimenter can turn the stress on and off, on the same sample, by adding the complementary DNA oligomer (cDNA) to a solution of ss chimeras, or displacing the cDNA from the chimera by adding a competitor. Another, irreversible way to turn the stress off is to cut the DNA spring with a restriction enzyme. Moreover, different states of stress can be achieved with the same chimeric construct by partially hybridizing the DNA spring, using complementary oligomers of different lengths (Fig. 1D). The external control allows, on one hand, to perform quantitative experiments on the effect of mechanical stress on enzyme function, and, on the other hand, is the basis for turning the enzyme-DNA chimeras into molecular probes.

2.2 Examples of different enzymes modulated by the enzyme-DNA method

The enzyme-DNA method has been successfully utilized in modulating several enzymes. Our first successful target, maltose binding protein (MBP), is actually a transport protein, not an enzyme; we mention it because historically it was the first protein with which the method of the DNA springs was tried out (Choi et al., 2005). Under stress, a modest decrease (about a factor 2) in the binding affinity for maltose was achieved. This construction used a 60 bp DNA spring. MBP is a classic example of induced fit binding; its general structure consists of two lobes, with the maltose binding pocket located in between. Upon binding of maltose, the two lobes of the protein

tilt towards each other, enclosing the maltose. The attachment points of the DNA spring were chosen so that the mechanical tension would work against the conformational motion driven by the binding of maltose (this consists of the two lobes of the molecule tilting closer to each other). On general thermodynamic grounds (Le Chatelier's principle), a perturbation which works against the conformational change induced by maltose binding must decrease the binding affinity for maltose. However, the method is far more general, as shown by the next example.

Guanylate Kinase (GK) is a 22 kDa monomeric enzyme that phosphorylates guanosine monophosphate (GMP). Binding of GMP induces a large conformational change similar to the MBP, in that the two lobes of the molecule “close” around the substrate (Fig. 1A). We have studied three different chimeras of this molecule, with the DNA spring attached as shown in Fig. 2A. In the presence and absence of mechanical stress, we measured the enzymatic speed as a function of substrates' concentration, and extracted the Michaelis-Menten (MM) constants $K_{M,G}$ (for GMP) and $K_{M,A}$ (for ATP), and the catalytic rate k_{cat} (Tseng, Wang, & Zocchi, 2010). The important result is that the three different configurations of the DNA spring (i.e. the three different states of internal stress applied to the enzyme) modulate the parameters $K_{M,G}$, $K_{M,A}$, k_{cat} in different ways (as opposed to modulating the parameters in the same way but to different degrees). Specifically, tension along the 40/171 direction decreases k_{cat} to about 40% of the original value, while $K_{M,G}$ and $K_{M,A}$ remain essentially unaffected. In contrast, tension along the 75/171 direction affects mainly $K_{M,G}$ (Fig. 2B). We conclude that the effect of mechanical stress is not simply to bias the structure towards one or the other of the two conformations (“open” and “closed”) seen in the X-ray structures, but is, in general, a stress specific deformation of the molecule. The DNA spring allows to access deformed states of the enzyme beyond the free and ligand bound states. Note that proteins are “soft” in the sense that thermal fluctuations around the average structure are relatively large. Fluctuations are also functionally important. In this context, a “deformed state” is a state where the average position of some atoms is different with respect to the reference state, or else a state where the fluctuations of some atoms around their average position is different compared to the reference state (Cooper & Dryden, 1984; Zocchi, 2018).

The next example is an enzyme that is not of the induced fit kind as it does not undergo apparent conformational change of the structure with and without substrates bound. Renilla Luciferase is a monomeric, 36 kDa enzyme that catalyzes the oxidation reaction of the luciferin coelenterazine,

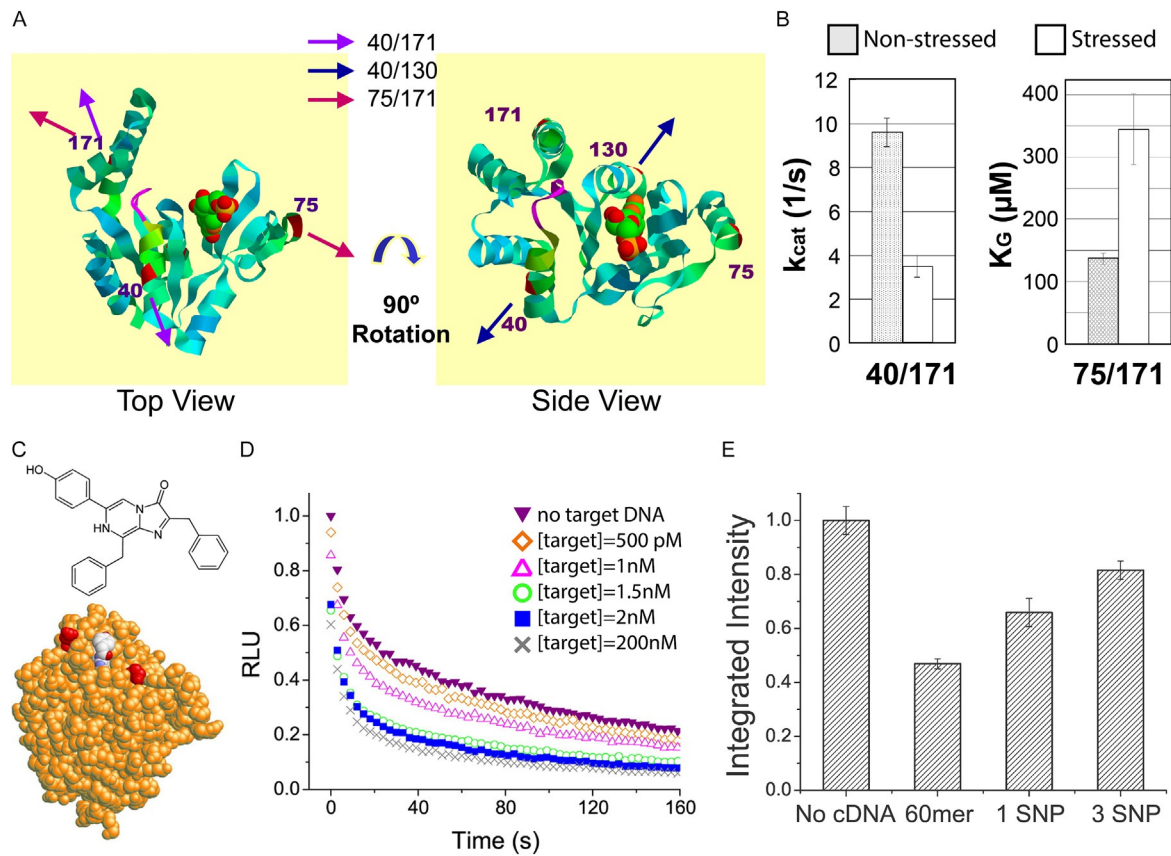


Fig. 2 See figure legend on next page.

using molecular oxygen dissolved in the medium. The reaction results in the emission of a blue photon (center wavelength is 470 nm, quantum efficiency is about 5%). Because of the light emission, this system is well suited as a platform to develop various molecular probes. Coelenterazine binds in a deep pocket in the enzyme (Figs. 1B and 2C), with no evident conformational change of the average structure with and without the ligand bound. Nonetheless, through a DNA spring attached as shown in Fig. 1B we could substantially modulate the enzymatic activity. With a 60 bp DNA spring we could dim the light emission to 36% of the original value (Tseng & Zocchi, 2013), and with a 40 bp spring it is reduced to 15% of the original value (Tseng & Zocchi, 2014). The applied stress does not simply unfold a fraction of the molecules. This is seen, for example, from the time course of the emitted light intensity under stress (Fig. 2D), which cannot be rescaled to coincide with the no-stress time course. We evince that the DNA spring induces a stress-specific deformation of the folded enzyme.

Fig. 2 (A) The figure shows three pairs of attachment points for the DNA spring on the enzyme Guanylate Kinase, which we used in the experiments. The GK structure (PDB: 1ZNX) is shown with bound GMP (space-filled). The three pairs of modified sites for DNA spring attachments (residue 40/171, 40/130, 75/171) are colored red. For each pair, the arrows illustrate the direction of the average mechanical force applied to the enzyme by the DNA spring in the “rigid” (ds) state. (B) The major effect of mechanical stress on the kinetic parameters measured for two of the GK-DNA chimeras described in (A). Shown is the catalytic rate k_{cat} for the 40/171 configuration, and the Michaelis-Menten constant relative to GMP for the 75/171 configuration, in the presence (white bars) and absence (gray bars) of mechanical stress. (C) Crystal structure of Renilla luciferase with bound substrate (coelenterazine-h, chemical formula shown) from Protein Data Bank (PDB) structure 2PSJ. Residues 161 and 188 (colored red) are mutated to cysteines in the experiments, serving as the attachment points for the DNA spring in the RLuc chimera. (D) The time course of luminescence intensity (arbitrary units) for the RLuc chimera from a titration experiment with varying amounts of complementary DNA (cDNA). The chimera DNA and the cDNA are both 60 bases long. From these data one can show that the fraction of hybridized chimeras is stoichiometric with the target cDNA down to the smallest concentration tested (500 pM) (Tseng & Zocchi, 2013), providing sensitivity in target detection. The chimera concentration was approximately 2 nM. (E) SNP detection with the RLuc chimera. By measuring the integrated intensity, the RLuc chimera we developed is able to detect one mismatch (1SNP) and three mismatches (3SNP) in the center of the DNA spring (60 bp). Panels (C–E) are reprinted with permission from Tseng, C.-Y., & Zocchi, G. (2013). *Mechanical control of Renilla luciferase*. *Journal of the American Chemical Society*, 135(32), 11879–11886. doi:10.1021/ja4043565. Copyright (2013) by American Chemical Society.



3. Principles of designing enzyme-DNA chimeras

3.1 Selection of the enzyme and DNA attachment sites

If we use the DNA spring as an experimental tool to study conformational plasticity in enzymes, the choice of enzyme is evidently dictated by the program of study. However, there may be other considerations and constraints. Our most successful constructions to date were with monomeric proteins in the 20 to 40 kDa range. With multimeric enzymes, obtaining functional enzyme-DNA chimeras may be more difficult, and require new protocols. Possible concerns include whether the DNA will interfere with the multimerization, whether the DNA should be coupled to the individual monomers or to the multimeric complex, and the yield of different species. There are also opportunities: for a dimeric enzyme complex, chimeras could be constructed with the DNA spring acting on one of the monomers, or the other, or across the two. Our only experience of controlling a multimeric complex is with the Protein Kinase A (PKA), which we will discuss in detail in [Section 5.1](#).

When planning a chimera construction, one of the first questions is of course where to attach the DNA spring on the surface of the enzyme in order to obtain control of the enzymatic activity. In our experience so far, virtually any configuration of the DNA spring likely will result in some effect on enzymatic activity, which is consistent with our view of the enzyme as a deformable viscoelastic solid. On the other hand, predicting where the DNA spring should be attached in order to obtain optimum control of the enzyme, is another matter, and we do not have an answer to this most interesting question. However, we may provide some suggestions. There have been attempts to identify “hot spots” for allosteric control on the surface of enzymes ([Halabi, Rivoire, Leibler, & Ranganathan, 2009](#); [Lockless & Ranganathan, 1999](#)) and experiments where a light sensitive conformational switch was inserted at these locations obtained some control ([Lee et al., 2008](#); [Reynolds, McLaughlin, & Ranganathan, 2011](#)). From our experience, using the DNA spring to mechanically control enzymes against their conformational changes during their catalytic cycles is a promising strategy, for example, in the cases of Maltose Binding Protein and Guanylate Kinase. Perturbing the substrate binding or the catalytic motif of enzymes is also demonstrated to work well, for example, in the cases of GK and Renilla Luciferase (see [Sections 2.2](#) and [5.1](#) for more details).

To construct the enzyme–DNA chimeras, it is desired to have exactly two cysteines on the surface of the enzyme, defining the DNA attachment points. These cysteines are generally introduced by mutagenesis; in doing so, the objective is to obtain reactive (i.e. exposed) Cys, and to avoid significantly perturbing the enzyme. For the latter requirement, one must verify *a posteriori* that the enzymatic activity of the protein is not too much affected by the mutagenesis. Our strategy was to identify from the crystal structure well exposed surface residues chemically similar to Cys—ideally Ser, but we also successfully substituted Thr, Glu, Arg, Lys—and substitute with Cys. Removing internal cysteines might also help to improve the success and yield of the desired enzyme–DNA chimeras, and remove any ambiguity in identifying correct constructions by gel electrophoresis or HPLC. For example, for the study corresponding to Fig. 2A (Tseng et al., 2010; Wang, Wang, Qu, & Zocchi, 2009) we substituted the two internal Cys of the enzyme with Ser, and the DNA attachment points are chosen away from the active site of the enzyme, so as not to interfere with enzyme function. It is worthwhile to note that if a native cysteine is internal and inaccessible to the crosslinkers, it may be unnecessary to replace the Cys with Ser. For example, we have been successful to construct functional chimeras with RLuc without removing the internal cysteines in the protein. Our approach is to conduct a preliminary test by trying to react the wild-type protein with excess amount of crosslinker-conjugated DNA (Cx-DNA) and inspect the products; if Cx-DNA do not attach to the proteins, i.e. no protein–Cx-DNA (one-arm chimera) or protein with multi DNA arms are formed, then Cysteine substitution is not required. Namely, even if a protein contains a structurally important Cysteine or disulfide bond, the method may still be applicable.

3.2 Design of the sequence of the DNA arms

It is suggested that according to the size of target enzyme, the length of the DNA arms generally falls in the range of 40 to 60 bases, which, upon hybridization and bending, introduce significant mechanical stress on the enzyme of interest. The question of the optimal length of the DNA spring is delicate; from a theoretical perspective, it touches on several fundamental but still imperfectly known aspects of protein (and DNA) mechanics; experimentally, it could certainly be (laboriously) determined by construction for any given enzyme and attachment sites (and cross-linkers). For a quantitative understanding we need a model, which is briefly discussed in Section 5.2.

For the interested readers, we refer to these previous publications (Qu, Wang, Tseng, & Zocchi, 2011; Qu & Zocchi, 2011). The sequence of DNA arms should be carefully designed to avoid hairpin formation, self-dimerization, and self-hybridization, which can be achieved by using online web servers (e.g., Nupack) or the analyzer tools provided from DNA manufacturers (e.g., OligoAnalyzer from Integrated DNA Technologies). The DNA arms should also be modified to have amino groups, which will be cross-linked to the cysteines on the surface of the enzyme of interest via bifunctional cross-linkers. One of the DNA arms should also be phosphate-modified to enable the ligation step of the construction. We provide the sequences of our DNA arms in [Section 4.2](#) as examples.

3.3 Two-step chimera construction

Enzyme-DNA chimeras are constructed in two sequential steps (see [Fig. 3](#) for a flowchart). In the first step, the first DNA arm is attached to one surface cysteine of the protein, forming a one-arm (1R) chimera, through a crosslinker. The crosslinker we utilize here is SM(PEG)₂, consisting of N-hydroxysuccinimide (NHS) ester and maleimide reactive group linked by a 17.6 Å PEG spacer. The NHS esters react with the amino modifications at the end of the DNA arms while the maleimides react with the sulfhydryl groups of cysteines on the protein surface. Since there are two reactive cysteines on the protein, unavoidably some two-arm chimeras (with identical arms) are also obtained, as well as protein dimers (bound by one di-sulfide bond) with one or two DNA arms; therefore, reduction by tris (2-carboxyethyl)phosphine (TCEP) and purification by HPLC with size-exclusion columns and ion-exchange columns are necessary after the first step. Size-exclusion column is not only used to filter out protein aggregation, but also to remove TCEP, which interferes DNA conjugation in 1R construction. We tested and found the maximum allowable concentration of TCEP is 10 μM (Wang et al., 2009) ([Fig. 4D](#)). For the second step in the synthesis, an excess amount of the second DNA arm reacts with the purified 1R, followed by Ni-NTA chromatography to remove uncoupled DNA. The purified two-arm (2R) chimeras are subsequently ligated with T4 DNA ligases, followed by Ni-NTA chromatography to obtain the final product, i.e., the ligated two-arm (L2R) chimeras. The two-step chimera construction is superior to one-step method, in which a single DNA strand is attached on the protein surface at both ends in one step, for its much higher yield (see Wang et al., 2009 for detailed discussions).

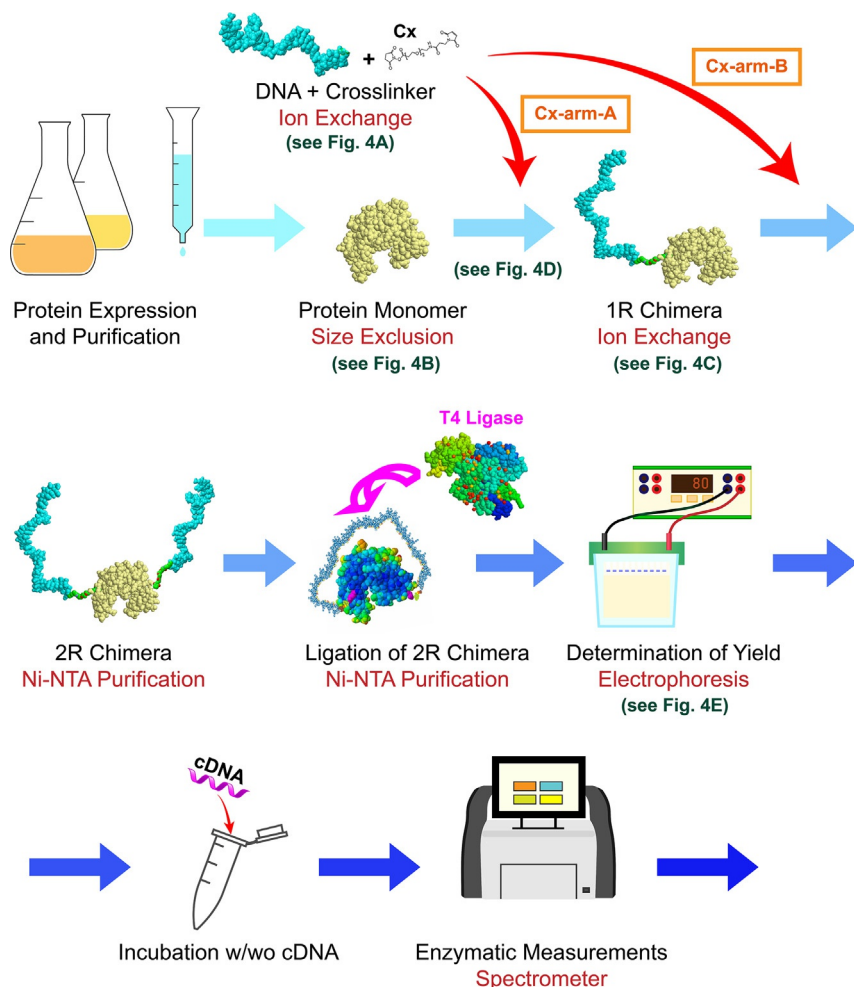


Fig. 3 Flowchart for synthesizing ligated two-arm chimeras. The step-by-step protocol is given in [Section 4](#). The corresponding HPLC traces or gel images of the steps are noted and can be referred to [Fig. 4](#).



4. Design and synthesize enzyme-DNA chimeras

4.1 Equipment

Protein expression system, high performance liquid chromatography (HPLC), centrifuge, gel electrophoresis system, spectrometer.

4.2 Chemicals and materials

Protein expression materials and reagents (plasmid, competent bacteria, bacteria growth media, isopropyl β -D-1-thiogalactopyranoside (IPTG), etc.), Ni-NTA purification reagents, cross-linker (e.g., SM(PEG)₂, an amine-to-sulphydryl crosslinker with 17.6 Å spacer arm, Thermo Scientific, Cat. No: 22102), Tris(2-carboxyethyl)phosphine (TCEP), T4 ligase.

- Synthesized DNA (with end modifications):

DNA arm-A: 5'-amino modification-sequence (20 ~ 30 nt)-3'

DNA arm-B: 5'-phosphate modification-sequence (20 ~ 30 nt)-amino modification-3'

DNA splint for ligation (12–18 nt)

Target cDNA: complementary DNA to arm-A plus arm-B

- Examples of DNA arms (from Integrated DNA Technologies):

arm-A (30 nt): 5'-/5AmMC6/GAGTGTGGAGCCTAGACCGT GAGTTGCTGG-3'

arm-B (30 nt): 5'-/Phos/CAGGAGTCCACGGTACCATCCAAG CAGCTG/3AmMO/-3'

Splint (18 nt): 5'-GGACTCCTG CCAGCAACT-3'

Target cDNA: 5'-CAGCTGCTTGGATGGTACCGTGGACTC CTGCCAGCAACTCACGGTCTAGGCTCCACACTC-3'

Note: 5AmMC6 and 3AmMO are amino-modifications positioned at the 5'-end and 3'-end, respectively, of synthetic DNA provided by Integrated DNA Technologies (IDT).

- Buffers for HPLC:

Low salt: 100 mM sodium phosphate, and 1 mM EDTA at pH 7.

High salt: 100 mM sodium phosphate, 1 M NaCl, and 1 mM EDTA at pH 7.

4.3 Protocols

4.3.1 Two-arm chimera (DNA-protein-DNA) synthesis

1. Protein expression and purification

- Refer to other protocols for protein expression and purification.
- The protein of interest should contain a His-tag or another affinity-based purification tag, which will facilitate the purification using affinity purification protocols.

2. DNA-crosslinker conjugation and purification

- Dissolve DNA arm-A in phosphate buffer at pH 7.2–7.5 to prepare 1 mM stock and store at –20 °C.

- Dissolve crosslinker (Cx) SM(PEG)₂ in dimethyl sulfoxide (DMSO) to prepare 1 M stock solution, which is then divided into small aliquots and stored at -20°C . Freshly prepared Cx solution (<2 months) is preferred.
 - Mix Cx and DNA solution at the ratio of 50 (or more):1 in phosphate buffer at pH 7.2–7.5 and incubate for 1 h at RT.
 - Remove excess crosslinkers by using ion-exchange chromatography (e.g. UNO Q1 Column, Bio-Rad, Cat. No: 7200001; Fig. 4A). The pH of the HPLC collection should be adjusted to ~ 7 to facilitate the following 1R conjugation. Measure the Cx-DNA concentration. The yield of Cx-DNA can be determined by using 15% TBE-Urea gel (Fig. 4D). The Cx-DNA conjugate should be used freshly (<1 week stored at 4°C).
 - Repeat the above steps for DNA arm-B.
3. Protein monomer purification
- If the target proteins form dimers or polymers through di-sulfide bond between cysteines exposed on the surfaces (e.g. GK), perform the following steps; otherwise, skip this part (e.g. RLuc).
 - Add 1–5 mM TCEP to the protein solution at a final concentration of 1–5 mM and incubate for 30 min–1 h.
 - Remove excess TCEP using size-exclusion chromatography (e.g. Bio-Sil SEC 125-5 Column, Bio-Rad, Cat. No: 125-0060; Fig. 4B). The pH of the HPLC collection should be adjusted to ~ 7 for later one-arm (1R) conjugation. Measure the protein concentration. Perform 1R conjugation immediately to minimize protein re-polymerization.
4. One-arm (DNA-protein) and two-arm (DNA-protein-DNA) chimera synthesis
- Add Cx-arm-A to protein monomer and incubate for 3–5 h at RT. The ratio of Cx-DNA to protein depends on the properties of the protein and should be optimized to achieve higher yield of 1R constructions compared to 2R constructions. The optimized ratios vary from protein to protein, or even to protein with different attachment sites. For example, for GK we used 1:2 while for RLuc we used 1:9.
 - Purify the 1R chimera using ion-exchange chromatography (Fig. 4C) and measure the concentration.
 - Add Cx-arm-B to the purified 1R chimera at the ratio higher than 2:1 and incubate at 4°C overnight to maximize the yield.

5. Two-arm chimera purification

- Purify 2R chimera by Ni-NTA affinity purification procedures. Note that, due to the excess use of Cx-arm-B, the amount of 1R chimeras after the conjugation is significantly low compared to the desired 2R chimeras. The main goal of this step is to remove the DNA; therefore, Ni-NTA affinity purification is simple yet sufficient to achieve the purpose, while purification based on chromatography is expected to work well. Also, note that as the conjugation buffer contains EDTA, which interfere with the Ni-NTA affinity, the 2R chimera should be exchanged into other buffers using centrifugal concentrators before purification. However, certain enzymes (e.g., RLuc) are unstable and vortexing/centrifuging should be avoided at all times. In these cases, dilute the conjugation reaction by adding PBS buffer (pH 7.4) instead.

4.3.2 Ligation of two-arm chimera

1. Ligation and purification of two-arm chimera

- Mix 2 μM 18mer DNA splint, 2 μM two-arm chimera, 4 unit/ μL T4 ligase and 0.4 mg/mL BSA in recommended reaction buffer and incubate at 16 °C overnight. If the concentration of 2R chimera is low, optimize the concentrations of the reagents and the incubation time (e.g., 1.25 μM DNA splint, 20 nM chimera, 1.2 unit/ μL ligase and 0.1 mg/mL BSA were used for RLuc)
- To purify the ligated 2R chimera, follow the steps for purifying 2R chimera in Protocol 4.3.1

2. Determination of yield

- Perform SDS-PAGE electrophoresis to determine the yield of L2R chimera. Molecules with different molecular weight as well as different configurations (e.g. linear and circular shape) can be separated by using gel electrophoresis (Fig. 4E).

4.3.3 Enzymatic activity measurements

1. Measurements on effects by mechanical stress

- For the sample: add target DNA (complementary DNA of the arm A+B, i.e. cDNA) to the chimera in the reaction buffer with ~ 0.4 mg/mL BSA and incubate for 3 h to overnight. Generally, 5:1 of cDNA:chimera is sufficient to saturate the enzymatic reactions

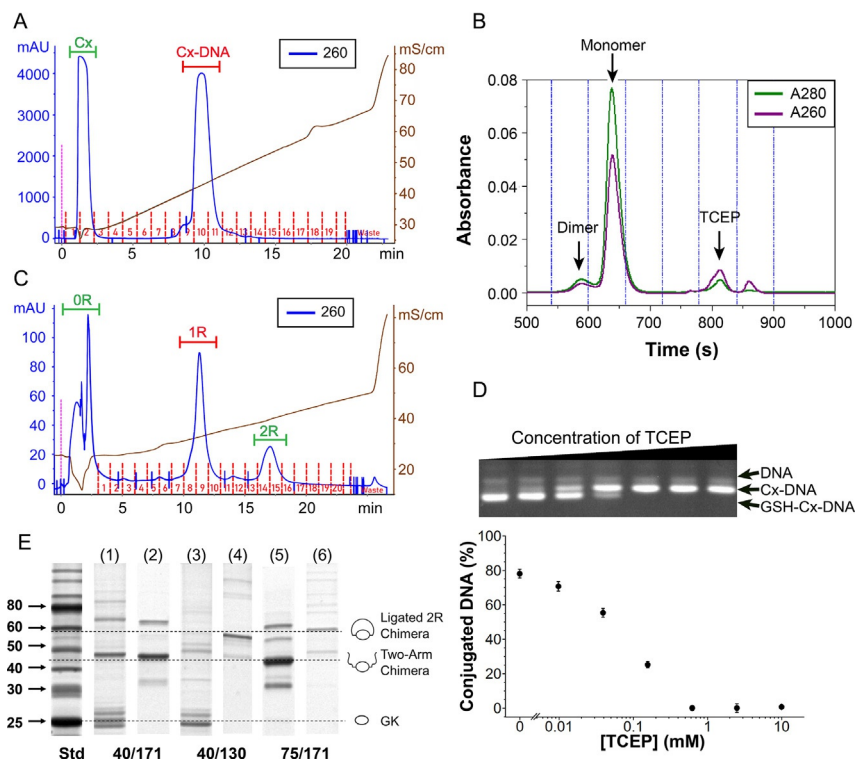


Fig. 4 Representative examples of preparative HPLC traces and gels at various stages during the synthesis of ligated two-arm chimeras. (A) High performance ion-exchange chromatography (IE-HPLC) profile for removing excess cross-linkers (SM(PEG)₂) from Cx-DNA. The blue curve is the absorption at 260 nm; the brown curve represents the conductivity. Red dashed lines and numbers indicate collection fractions. (B) Size-exclusion HPLC trace in terms of absorptions at 280 nm (green line) and 260 nm (purple line) of the purification step for removing TCEP from the reduced proteins (GK) prior to conjugation to DNA arms. Blue dash-dotted lines indicate different collection fractions. (C) IE-HPLC profile of 1R chimera purification (RLuc). The blue curve is the absorption at 260 nm; the brown curve represents the conductivity. Red dashed lines and numbers indicate collection fractions. (D) Verification of conjugation of crosslinkers (Cx) to DNA arms and conjugation of Cx-DNA to thiol groups in the presence of TCEP at different concentrations. Top: representative TBE-urea PAGE gel showing bands for unconjugated DNA (DNA), crosslinker-conjugated DNA (Cx-DNA), and glutathione conjugated Cx-DNA (GSH-Cx-DNA). Bottom: the dependence of the percentage of successfully conjugated GSH-Cx-DNA on the concentration of TCEP in the conjugation solution. These measurements give an estimate for the maximum residual concentration of TCEP which can be tolerated in the DNA-protein conjugation reaction. TBE-Urea gel can also be used to determine the crosslinking efficiency of the DNA-arms and the cross-linkers. (E) Gel electrophoresis of products after protein, 2R chimera and L2R chimera purification (GK). The quantities of the desired products and side products can be determined

(Continued)

- For the control: add nothing (if non-mechanical effects are absent, see [Section 4.5.3](#)) or two complementary DNA strands (one for arm-A and the other for arm-B) to the chimera in the reaction buffer with ~ 0.4 mg/mL BSA and incubate for 3 h to overnight. The cDNA:chimera ratio should be the same as the sample
- Perform enzymatic activity measurements for the sample and control side by side

4.4 Example of protocol—RLuc-DNA chimera synthesis

In addition to the general protocols ([Section 4.3](#)), here we provide the exact steps that we used to synthesize RLuc-DNA chimeras and the readers may follow directly.

1. Express Renilla Luciferase mutant (cysteine mutations at 161, 188) in *E. coli* BL21(DE3)pLysS and purify it using Ni-NTA chromatography. The final concentration is ~ 20 – 30 μ M.
2. Dissolve 100 mg SM(PEG)₂ crosslinker (Thermo Scientific, 22102) (Cx) in 235 μ L DMSO to a final concentration 1 M. Aliquot to 10 μ L in 0.2 mL PCR tubes and store at -20 $^{\circ}$ C.
3. Dissolve customized DNA arm-A and arm-B (IDT) in 100 mM sodium phosphate buffer (pH 7.4) to a final concentration 1 mM. Store at -20 $^{\circ}$ C.
4. Mix 70 μ L of arm-A, 10 μ L of Cx with 100 μ L phosphate buffer (pH 7.4) and incubate for 1 h. Inject the mixture onto HPLC ion-exchange column (Uno Q1 column, Bio-Rad, Cat. No: 7200001) and collect the Cx-arm-A fraction. The final concentration is ~ 50 μ M.

Fig. 4—Cont'd by gel electrophoresis. After protein purification, the desired product is protein monomers and the side products include dimers and other multimers. After 2R chimera purification, the side products include 1R chimera while after L2R chimera purification, the side products include unligated 2R chimera. Lanes 1 and 2: 40/171 protein and 2R chimera purification; lanes 3 and 4: 40/130 protein and L2R purification; lanes 5 and 6: 75/171 2R and L2R purification. The small icons indicate the locations of protein monomers, 2R and L2R chimeras in the gel. Side products (e.g. dimers after protein purification and 1R chimeras after 2R chimera purification) are unavoidable. *Panel (D)* Modified with permission of IOP Publishing, Ltd., from Wang, Y., Wang, A., Qu, H., & Zocchi, G. (2009). Protein–DNA chimeras: Synthesis of two-arm chimeras and non-mechanical effects of the DNA spring. *Journal of Physics: Condensed Matter*, 21(33), 335103. doi:10.1088/0953-8984/21/33/335103; permission conveyed through Copyright Clearance Center. *Panel (E)* Modified with permission from Tseng, C.-Y., Wang, A., & Zocchi, G. (2010). Mechano-chemistry of the enzyme guanylate kinase. *Europhysics Letters Association*, 91(1), 18005. doi:10.1209/0295-5075/91/18005). Copyright (2010) by European physical society.

5. Mix 65 μL of arm-B, 10 μL of Cx with 100 μL phosphate buffer (pH 7.4) and incubate for 1 h. Inject the mixture onto HPLC ion-exchange column and collect the Cx-arm-B fraction. The final concentration is $\sim 40 \mu\text{M}$.
6. Mix 1000 μL RLuc mutant ($\sim 30 \mu\text{M}$), 65 μL Cx-arm-A ($\sim 50 \mu\text{M}$) with 900 μL phosphate buffer (pH 7) and incubate for 3 h. Inject the mixture onto HPLC ion-exchange column and collect the 1R chimera fraction.
7. Add 150 μL of Cx-arm-B into the collected fraction.
8. Repeat step 6 and 7 for three times to collect enough product.
9. Incubate the product at 4 $^{\circ}\text{C}$ overnight.
10. Mix 600 μL Ni-NTA beads, the product of step 9, with PBS buffer to a total volume $\sim 50 \text{ mL}$. Incubate the mixture on shaker for 4 h at 4 $^{\circ}\text{C}$.
11. Pass the mixture through gravity flow column. Discard the flow-through, wash the beads with 10 mL of PBS buffer, elute the product with 500 μL of elution buffer (containing 100 mM imidazole) for four times and collect each elution in individual tubes. Measure the luminescence intensity of each collection and keep the ones with high chimera concentration.
12. Determine the yield of successful construction (2R chimera) with 10% SDS-PAGE gel.
13. Add 100 μL of 10X ligation buffer, 10 μL of 100 μM DNA splint, 30 μL of BSA (20 mg/mL), 300 μL product of step 11, 20 μL of T4 ligase (1–3 unit/ μL) with 540 μL of ultra pure water in a 1.5 mL tube and incubate at 16 $^{\circ}\text{C}$ overnight (or longer). We usually prepare three sets of samples to collect adequate quantity.
14. Mix 500 μL Ni-NTA beads, the product of step 13 with PBS buffer to a total volume $\sim 50 \text{ mL}$. Incubate the mixture on shaker for 4 h at 4 $^{\circ}\text{C}$.
15. Pass the mixture through gravity flow column. Discard the flow-through, wash the beads with 3 mL of PBS buffer, elute the product with 500 μL of elution buffer for four times and collect each elution in individual tubes. Measure the luminescence intensity of each collection and keep the ones with high chimera concentration.
16. Determine the yield of successful construction (Ligated 2R chimera) with 10% SDS-PAGE gel.
17. Incubate the L2R chimera with/without cDNA for 1 h (to overnight) and measure the luminescence activity.

4.5 Troubleshooting

4.5.1 Protein aggregation

Protein aggregation may be induced due to various reasons. Cysteine mutations introduced on the protein surface may lead to aggregation via the formation of disulfide bonds, for example, in the case of GK. We use TCEP to reduce the disulfide bonds and obtain protein monomers (Fig. 4B). The concentrations of TCEP and incubation time could be optimized. Reduced proteins will begin to polymerize again once TCEP is removed; therefore, it is strongly suggested that 1R conjugation should follow immediately the removal of TCEP. In addition, some proteins tend to aggregate (e.g. RLuc). Our strategy for this situation is to keep the protein concentration low ($\sim 25 \mu\text{M}$ for RLuc) and avoid vortexing at all times.

4.5.2 Low yield of 1R chimera

One-arm chimera is generally the most critical step and may require significant effort of optimization. Low yield of 1R chimera may be due to the failure in Cx/DNA or Cx-DNA/protein conjugation, or the formation of side products. Cross-linkers are highly sensitive to pH and storage time. If cross-linkers fail to conjugate to either DNA-arms or proteins, we suggest checking the pH at each step and using fresh cross-linkers (< 2 months) and Cx-DNA (< 1 week). The side products, including 2R chimeras (with identical arms) and dimers with one or two DNA arm, are unavoidable but can be reduced. If there is a large quantity of 2R chimera, we suggest increasing the protein/DNA ratio until the percentage of 1R chimera is satisfactory. If there are large quantities of dimer products, a higher concentration of TCEP or incubation time before 1R purification may be helpful.

4.5.3 Non-mechanical effect of DNA on enzymatic activity

Certain enzymes or proteins may suffer from non-mechanical effect due to the conjugation to DNA. Examples of such proteins include DNA-binding proteins (e.g., *lac* repressor) and proteins interacting with substrates containing the pyrimidine and purine bases (e.g., GK interacting with ATP and GMP). In these cases, an activation of the enzymatic activity could be resulted from the removal of DNA from the substrate binding domain rather than mechanical stress. To correctly evaluate the influence of the mechanical stress, special care should be taken by proper controls (Wang et al., 2009). For example, one can compare the enzymatic activity of the chimera hybridized with one cDNA, which forms an intact dsDNA spring

and exerts mechanic stresses, with that of the chimera hybridized with two individual cDNA, which leaves nicks or gaps in the dsDNA and thus does not stress the enzyme.



5. Discussion

5.1 Selection of target enzyme: Possibilities and challenges

Our first chimera constructions were with Maltose Binding Protein and Guanylate Kinase; both proteins undergo large conformational changes induced by ligand binding, and the attachment of the DNA spring was initially designed to oppose part of that conformational change (e.g. the 75/171 attachments in Fig. 2A), and thus affect ligand binding. These designs were successful in mechanically controlling both proteins. In another case (the 40/171 attachments in Fig. 2A) we purposely tried to perturb elements of the catalytic motif (the so-called p-loop in GK), which also resulted in successful mechanical control of the enzyme. For the RLuc, we sought to perturb the substrate binding pocket (Figs. 1B and 2C), and obtained mechanical control even though substrate binding is not of the induced fit type.

So far, our only attempt to control a multimeric complex was with the Protein Kinase A (PKA). This allosterically controlled enzyme consists of a catalytic and a regulatory subunit; binding of cAMP to the regulatory subunit causes it to fall off the complex, exposing the catalytic site and thus activating the enzyme. We attached a DNA spring to the regulatory subunit, in a position such that the mechanical stress would favor part of the documented conformational change upon cAMP binding. We were able to activate the enzyme under stress, though the effect was about half of that obtained with cAMP (Choi & Zocchi, 2006). However, at the time we were using a different chimera synthesis method (coupling to the enzyme a single DNA oligomer modified at both ends), which resulted in a lower yield of correct constructions in the experiments. An interesting question is whether one can even improve on the natural, cAMP driven allosteric regulation (in terms of the ratio of enzymatic activity in the on and off states). There is a trade-off between the residual activity in the off state and the activity in the on state. Increasing the bond strength between the regulatory subunit (which acts as an inhibitor) and the catalytic subunit would decrease the residual activity in the off state, but also decrease the fraction of dissociated enzyme complexes (and thus the enzymatic activity) in the on state.

If one could engineer a stronger bond between the two subunits, and if the mechanical stress provided by the DNA spring were large enough to still drive almost complete dissociation of the subunits, one might increase the dynamic range of allosteric control.

5.2 Quantitative characterization of the mechanical perturbation

In [Section 3.1](#) we noted that, in our experience, a mechanical perturbation at virtually any location on the enzyme will have some effect on the activity. However, obtaining a measurable effect must also depend on the strength of the perturbation. Therefore it would be desirable to have a measure to quantify this strength in mechanical terms, that is, in terms other than the effect on enzymatic activity. We now summarize our progress in this direction.

A complete microscopic characterization would be in terms of the stress field in the enzyme; at the other extreme, if one wished to characterize the perturbation with just one number, it would be the total elastic free energy injected into the enzyme, E_{enzyme} (in the following we will call it “elastic energy” for short). It is not easy to measure this quantity. We were able to measure a related quantity: the total elastic energy in the enzyme-DNA chimera under stress (i.e. the sum of the elastic energies in the enzyme and the DNA spring). These measurements are detailed here ([Tseng, Wang, Zocchi, Rolih, & Levine, 2009](#)), and an overview is also given here ([Zocchi, 2018](#)). The method we used exploits the elastic energy driven dimerization of certain enzyme-DNA chimeras (namely, those constructions with a nick in the DNA spring). With the same method, we measured the elastic energy of the DNA spring ([Qu & Zocchi, 2011](#)). For these cases we therefore know the elastic energy injected into the enzyme: $E_{enzyme} = E_{tot} - E_{DNA}$, where E_{tot} is the total elastic energy in the chimera, and E_{DNA} the elastic energy in the DNA spring. We note that it should also be possible to measure these elastic energies by calorimetry, though these would be quite delicate experiments. To give some actual numbers, for the GK-DNA chimera with a 60bp DNA spring attached at the sites 75/171 ([Fig. 2A](#)), we find $E_{tot} \approx 9.1 k_B T$, $E_{DNA} \approx 6.6 k_B T$, and thus $E_{enzyme} \approx 2.5 k_B T$ ([Zocchi, 2018](#)). Here $k_B T$ is the thermal energy at room temperature, $1 k_B T = 0.59$ kcal/mol.

To build an understanding of these numbers, and of the mechanical perturbation in general, one can build a model. The minimal such model considers the enzyme-DNA chimera as a system of two coupled springs: a DNA “leaf spring” which is compressed, and an “enzyme spring” which is stretched. We know the mechanical response of the DNA spring, from a

combination of measurements and model building (Qu & Zocchi, 2011). It is a nonlinear spring, characterized by a softening transition that limits the force it can deliver to a few pN. It turns out that the enzyme spring is also nonlinear, with a similar softening transition (Tseng & Zocchi, 2014). With this model of two coupled nonlinear springs we can rationalize the measurements on the enzyme–DNA chimeras, and make some predictions. Given the geometry of the chimera construct (distance between surface attachment points in the undeformed enzyme, contour length of the cross-linkers, contour length of the DNA spring), and assuming a specific linear and nonlinear mechanical response of the enzyme, one can calculate the elastic energy injected in the enzyme (the energy in the “enzyme spring”), as well as other thermodynamic quantities. If we assume that enzymatic activity is a monotonically decreasing function of the elastic energy in the enzyme, we can then predict the effect of changing the length of the DNA spring (or other parameters) on the dynamic range of control obtained in the corresponding chimeras. In this way we understand, for example, why a 40 bp DNA spring elicits stronger control of the enzyme compared to a 60 bp spring, for the chimera of Fig. 1B. More details are provided in (Tseng & Zocchi, 2014).

5.3 Applications: Enzyme–DNA chimeras as molecular probes

Sensitive molecular detection systems often use an enzyme based amplification mechanism, as in enzyme-linked immunosorbent assay (ELISA) or polymerase chain reaction (PCR). An enzyme–DNA chimera does in principle constitute a detection system for the DNA (or RNA) sequence complementary to the chimera DNA, and more generally for any binding event or process which modifies the stiffness of the DNA spring. The RLuc–DNA chimera in particular—as well as future constructions utilizing any of the enzymes commonly used in ELISA—lends itself to develop in-solution (homogeneous) molecular probes. We have only taken the first steps in this direction, with a proof-of-concept of using the RLuc–DNA chimera for direct DNA sequence detection (Fig. 2D) and mismatch detection (Fig. 2E) in solution (Tseng & Zocchi, 2013). But there are wider possibilities. Namely, it turns out that, just as ligand binding to the enzyme is not impaired by the mere presence of the DNA spring, so is the DNA in the chimera still available for protein binding. For example, restriction enzymes can bind to the DNA spring and cut it, and the very construction of the chimeras uses ligation of the DNA spring. Our preliminary, unpublished results show that we can detect DNA binding proteins with the RLuc chimera. The mechanism is, to our present understanding, that binding of a protein

to the ds DNA spring rigidifies the spring, causing a modulation of activity of the enzyme. In a different twist, we have used the RLuc-DNA chimera to measure the bending stiffness of exotic nucleic acids hybrids such as PNA/DNA and LNA/DNA (Zhan & Zocchi, 2017).



6. Summary

Various DNA-based techniques have been used to study and modulate the activity of enzymes (Engelen, Janssen, & Merkx, 2016; Zhang, Nangreave, Liu, & Yan, 2014). Among them, the method of enzyme-DNA chimera provides a distinctive approach; it works in solution and the stress applied on the enzyme can be fine-tuned semi-continuously. We have successfully constructed several enzyme-DNA chimeras using different enzymes, DNA springs, or attachment points, demonstrating that this is indeed a general method in artificial control of enzymes. With DNA springs, we can induce various states of deformation of a protein with tunable, rather than merely on/off, forces. For enzymes, deforming the molecule results in a modulation of enzymatic activity, in which the enzymatic parameters may be affected individually. For those interested in the general field of conformational changes and allostery, the method represents a relatively new experimental tool. For applications, the modulation of enzymatic activity in principle offers a basis to design various in solution and possibly in vivo molecular probes.

Acknowledgments

C-YT acknowledges support from Dr. Chin-Lin Guo and Academia Sinica of Taiwan (Grant No. AS-105-TP-A04, AS-TP-109-M04). YW acknowledges support from the National Science Foundation (Grant No. 1826642) and the Arkansas Biosciences Institute (Grant No. 0326). GZ acknowledges support from the National Science Foundation (Grant No. DMR-1809381).

We wish to acknowledge the fundamental contributions of, and the pleasure of working with, our colleagues Brian Choi, Andrew Wang, Hao Qu, Daniel Sanchez, Collin Joseph, and Yilin Zhan.

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